THE TRANSPORT CONJECTURE BETWEEN LATTICE SU(N) GLUEBALLS AND BIANCHI ORBIFOLD SPECTRAL OBSERVABLES: FORMAL STATEMENT AND FOUR CANDIDATE PROOF STRATEGIES

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ABSTRACT. We formulate explicitly an open conjectural identification, named the *Transport Conjecture*, between (i) the lattice SU(N) scalar glueball mass $m_{0^{++}}(\mathrm{SU}(N))$, and (ii) a spectral observable $m^{(\chi_*)}_{\mathrm{spectral}}$ on a Bianchi 3-orbifold Y_K with $K = K_{\mathrm{ASP}}(N)$ a privileged imaginary quadratic anchor. The conjecture is needed as an explicit substep in a surrogate mass-gap formula proposed in the companion paper [RŹ6c]; the present preprint isolates the conjecture as an *open problem* (TIER 3 in our internal classification), gives its explicit formal statement, and lists four candidate proof strategies (CCHS 4D extension; Cao–Sheffield random surfaces; Wightman positivity W0; Cluster decomposition W5). We also state five falsifiable per-anchor predictions, and conclude with an honest assessment of the open-problem status.

1. THE TRANSPORT CONJECTURE

1.1. **Setup.** Let $N \geq 2$ be an integer and let $\mathbb{T}^4 = (\mathbb{R}/L\mathbb{Z})^4$ be the 4-torus of edge L. Let $\mathcal{A}_{\mathrm{SU}(N)}(\mathbb{T}^4)$ denote the lattice SU(N) Yang–Mills theory on $\mathbb{T}^4$ with Wilson action (or any of its standard improvements), and let $\mathcal{O}^{(0^{++})} : \mathcal{A}_{\mathrm{SU}(N)}(\mathbb{T}^4) \to \mathbb{R}$ denote the lowest scalar glueball observable. In the continuum limit (lattice spacing $a \to 0$ at fixed physical volume), one defines

$$m^{\mathrm{lat}}_{0^{++}}(\mathrm{SU}(N)) \;:=\; \lim_{a\to 0} \sqrt{a} \cdot \left\langle \mathcal{O}^{(0^{++})}\, \hat{\mathcal{O}}^{(0^{++})} \right\rangle_a^{1/2}, \tag{1}$$

where the $\hat{}$ denotes a temporal-correlator extraction and $\langle\cdot\rangle_a$ the lattice ensemble average at spacing a.

On the arithmetic side, let $K = \mathbb{Q}(\sqrt{D})$ be imaginary quadratic with fundamental discriminant $D < 0$, $\mathcal{O}_K$ its ring of integers, $Y_K := \mathrm{PSL}_2(\mathcal{O}_K)\backslash\mathbb{H}^3$ the associated Bianchi 3-orbifold [EGM98, §1.5], and Δ the positive hyperbolic Laplacian on Y_K. By the per-character spectral decomposition of the companion paper [RŹ6b],

$$L^2(Y_K) = \bigoplus_{\chi \in \widehat{g}(K)} L^2(Y_K)_\chi,$$

and $\Delta_\chi := \Delta\big|_{L^2(Y_K)_\chi}$ has its own discrete cuspidal + continuous Eisenstein spectrum. The Center–Rank theorem CR of [RŹ6a] selects a privileged genus character χ_* for the anchor $K_{\mathrm{ASP}}(N)$ (the "central" character compatible with the 2-part of $Z(\mathrm{SU}(N))$). Define

$$m^{(\chi_*)}_{\mathrm{spectral}}(K, N) \;:=\; \sqrt{\lambda_1(\Delta_{\chi_*})}\,, \tag{2}$$

the square root of the lowest cuspidal eigenvalue of Δ_{χ_*} on Y_K.

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1.2. **Conjecture.**

Conjecture 1.1 (Transport Conjecture)**.** *Let the universal constants $\sqrt{2\pi e}$, $\sqrt{2/3}$ and the Dijkgraaf–Witten factor $F(N) = (9/10)(N^2+1)/N^2$ be as in* [Ŕ26c]. *Let*

$$\mathcal{C}(N) \;:=\; \sqrt{2\pi e}\cdot\sqrt{\tfrac{2}{3}}\cdot F(N)\,.$$

There exists a Bianchi orbifold Y_K with $K = K_{\mathrm{ASP}}(N)$ (in the sense of the Anchor Selection Principle of [Ŕ26a]*) such that, in the lattice continuum limit,*

$$\frac{m^{\mathrm{lat}}_{0^{++}}(\mathrm{SU}(N))}{\sqrt{\sigma_N}} \;=\; \mathcal{C}(N)\,, \tag{3}$$

where $\sqrt{\sigma_N}$ is the SU(N) string tension (extracted from the lattice via the standard Wilson loop area-law, or by any equivalent prescription).

Equivalently, there exists a transport map $\Phi : \mathcal{O}^{(0^{++})}|_{\mathbb{T}^4} \mapsto m^{(\chi_*)}_{\mathrm{spectral}}(K_{\mathrm{ASP}}(N), N)$ *such that the right-hand side of* (3) *factorises through the spectral observable on Y_K.*

1.3. **Numerical status (2026).** The Athenodorou–Teper 2021 lattice continuum data [AT21, Tab. 34] for SU(N), $N \in \{2,3,4,5,6,\infty\}$, gives an RMS deviation of 0.7% from the conjectural value $\mathcal{C}(N)$ defined in (3); see Table 1 of [Ŕ26c]. This numerical match is consistent with, but does not prove, the conjecture.

2. Four candidate proof strategies

We list four candidate strategies for promoting Conjecture 1.1 from TIER 3 (explicit open conjecture, with 0.7% RMS empirical match) to TIER 1 (rigorous proof). Each strategy is treated as a programmatic outline, not a road-mapped proof.

Strategy 1 (CCHS 4D extension)**.** The recent work of Chatterjee, Cao, Hairer, and Shen ([Cha16] and follow-ups; the 4D extension is, to the author's knowledge, still in progress as of mid-2026) constructs lattice SU(N) Wightman observables in 2D and 3D via regularity-structures techniques. A 4D extension, when it exists, would provide a rigorous existence theorem for $m^{\mathrm{lat}}_{0^{++}}(\mathrm{SU}(N))$ as a Wightman 2-point function correlation. The proof of Conjecture 1.1 via this route would then require: (a) the 4D CCHS construction; (b) identification of the Wightman 2-point function with the spectral observable on a Bianchi orbifold via a Bianchi-cycle gauge fixing.

Status (2026): CCHS-3D published; CCHS-4D in progress (no precise prediction by the author for completion date). Bianchi-cycle identification: open. Pessimistic 15-year estimate: 10-20%.

Strategy 2 (Cao–Sheffield random surfaces)**.** A recent programme by Cao and Sheffield (and collaborators) constructs random surface (Liouville) models that, in a certain limit, factor through gauge-theory observables on $\mathbb{T}^4$. If this construction extends to Bianchi 3-orbifolds (a manifold without flat 4D structure), it could provide a Bianchi-to-lattice transport: the Wilson loops on $\mathbb{T}^4$ would arise as random-surface measure averages, and the Bianchi-orbifold spectral observables would be obtained as a homotopy quotient of the Liouville model.

Status (2026): 2D Cao–Sheffield programme partly published; extension to 4D and to non-compact Bianchi 3-orbifolds is open.

Strategy 3 (Wightman positivity (W0))**.** Direct Wightman positivity [SW64]: prove that the lattice 2-point function in (1) is the spectral measure of an appropriate self-adjoint operator on a Hilbert space, and that this operator coincides with Δ_{χ_*} on $L^2(Y_K)_{\chi_*}$ after identification of the gauge-invariant Hilbert space with the Bianchi spectral Hilbert space. This is, in essence, the constructive QFT roadmap for Yang–Mills mass gap, restricted to the surrogate observable.

Status (2026): The full Wightman programme for 4D Yang–Mills is an open Clay Millennium problem [JW00]. A restricted version — proving Wightman positivity *only* for the surrogate observable, not for the full gauge-invariant ensemble — might be more tractable; this is the author's own preferred long-term route.

Strategy 4 (Cluster decomposition (W5))**.** Cluster decomposition in the spirit of [Wig56], [HK64]: prove that the lattice 2-point function clusters at large separations, hence its spectral support has a gap. If the spectral gap of the lattice theory coincides with the genus character spectral gap of Δ_{χ_*} on Y_K, the Transport Conjecture would follow at the level of spectral support (though not yet at the level of specific eigenvalue).

Status (2026): Cluster decomposition for lattice $\mathrm{SU}(N)$ is folklore-rigorous; the identification with Bianchi spectral support is open.

3. Five falsifiable predictions

The conjecture yields five testable predictions (cf. [R2́6c], §7); we restate them here, isolated, for ease of reference. Let $\mathcal{C}(N) := \sqrt{2\pi e} \cdot \sqrt{2/3} \cdot F(N)$.

F-T-1. $m^{\mathrm{lat}}_{0^{++}}(\mathrm{SU}(2))/\sqrt{\sigma_2} = \mathcal{C}(2) = 3.7962$. *Status (2026):* AT2021 reports 3.781 ± 0.023, $\Delta = -0.40\%$, PASS.

F-T-2. $m^{\mathrm{lat}}_{0^{++}}(\mathrm{SU}(3))/\sqrt{\sigma_3} = \mathcal{C}(3) = 3.3744$. *Status (2026):* AT2021 reports 3.405 ± 0.021, $\Delta = +0.91\%$, PASS.

F-T-3. $m^{\mathrm{lat}}_{0^{++}}(\mathrm{SU}(4))/\sqrt{\sigma_4} = \mathcal{C}(4) = 3.2273$. *Status (2026):* AT2021 reports 3.271 ± 0.027, $\Delta = +1.34\%$, PASS.

F-T-4. $m^{\mathrm{lat}}_{0^{++}}(\mathrm{SU}(5))/\sqrt{\sigma_5} = \mathcal{C}(5) = 3.1584$. *Status (2026):* AT2021 reports 3.156 ± 0.031, $\Delta = -0.06\%$, PASS.

F-T-5. $m^{\mathrm{lat}}_{0^{++}}(\mathrm{SU}(6))/\sqrt{\sigma_6} = \mathcal{C}(6) = 3.1213$. *Status (2026):* AT2021 reports 3.102 ± 0.032, $\Delta = -0.61\%$, PASS.

The RMS deviation across F-T-1–5 is 0.7%, comfortably inside the AT2021 continuum error bars $\sim 3\%$. The empirical evidence is suggestive but not conclusive: no datapoint deviates by more than the 1σ AT2021 error bar from $\mathcal{C}(N)$. Falsification of any one of F-T-1–5 by future high-precision lattice data would falsify the conjecture.

4. Honest open-problem status

The Transport Conjecture is an *open problem*. Its current status is summarised as:

- **Empirically:** consistent with all known lattice data (AT2021, 0.7% RMS over 6 datapoints).
- **Logically:** an explicit identification — not derivable from the heat-kernel + DW factors alone. Substitutes a non-trivial assumption into the surrogate mass-gap formula of [R2́6c].
- **Strategically:** all four candidate proof strategies in §2 are themselves open programmes (CCHS-4D in progress; Cao–Sheffield 2D-only; Wightman positivity is the Clay Millennium problem; Cluster decomposition is folklore but with an open Bianchi identification).

We invite the community to attempt the conjecture via any of these (or new) strategies.

4.1. What this paper does *not* claim.

- No proof. The conjecture is stated as an open problem.
- No claim of solution to the Yang–Mills Clay millennium problem. The conjecture is concerned with a specific surrogate observable.
- No claim that the RMS 0.7% empirical match is a proof. Numerical agreement, however suggestive, is not equivalent to a structural argument.

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